DE–6739 **11**

DISTANCE EDUCATION

P.G.D.C.A.(S) EXAMINATION, DECEMBER 2013.

First Semester

DIGITAL COMPUTER ORGANIZATION

(2011 Calendar year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. Define complements. Add using $r's$ and $(r-1)'s$ complement.

   (a) $(5438)_{10} + (-7432)_{10}$

   (b) $(10111101)_2 - (11011111)_2$.

2. Convert the following :

   (a) $(777.77)_{16} \rightarrow ()_{10}, ()_2$

   (b) $(1000000011)_2 \rightarrow ()_{10}, ()_8, ()_{16}$.

3. Simplify :

   (a) $F = x'yz + xyz + x \cdot x' + xy' + xy'z + xz$

   (b) $F = (x+y+z)(x'+y')(x'+y'+z')(x+y)$. Find SOP and simplify.

4. Design the 3×8 decoder circuit.

5. Design the full adder circuit.

6. Discuss on magnetic disk memories.

7. Write a short note on : Execution cycle.

8. Write short notes on programmed I/O.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Elucidate on number codes and character codes.

10. Simplify using K map and draw the circuit.

(a) $F(A,B,C,D) = \sum 0,1,2,3,4,6,8,10,11$

(b) $F(W,X,Y,Z) = \sum 2,4,6,8,9,10,11,12,$

dont care = $\sum 0,1,3,5,13$

11. Design the 8×1 MUX.

12. Design the parallel binary adder circuit.

13. Design the Master Slave Flip Flops and explain the concept with truth table.

14. Explain the working of BCD counter with the circuit.

15. Explain with the block diagram the working of priority interrupt.

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DE–6740 **12**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

First Semester

C AND DATA STRUCTURES

(2011 Calendar Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. How constants and identifiers are defined and used in C?

2. Explain the usage of Break and Continue statements.

3. How to pass the arguments to a function?

4. What is meant by Recursion? Explain with examples.

5. How to pass pointers to a function?

6. Discuss the concept of dynamic memory allocation.

7. Give the importance of structuring the data.

8. What is a Tree? Discuss on any one of its applications.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the various data types available in C language.

10. Write a program in C to calculate the roots of the quadratic equation $ax^2 + bx + c = 0$ where $a \neq 0$.

11. Explain the following in C :
    (a) MACROS
    (b) PREPROCESSOR

12. Write a program in C to create a Random file.

13. Write a program in C to add and subtract two matrices of order (3×3).

14. Explain the following operations in a stack :
    (a) PUSH and (b) POP

15. How to add and delete the nodes in a doubly linked list?

DE–6741 **13**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS

(2011 Calendar year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the purpose of primary key, and foreign key.

2. How to provide security for the database?

3. Discuss the data control facilities.

4. What are the different types of relations exist among the entities? Explain with examples.

5. Mention the properties of relational database.

6. Discuss the network system available in oracle.

7. How to work with NULL values in oracle?

8. Write a note on :

   (a) Renaming a table. (2)

   (b) Droping a table. (2)

   (c) Deleting a table. (2)

   (d) Truncating a table. (2)

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the concept of schema and subschema.

10. What is meant by database management system? Discuss its features and advantages.

11. Explain the grant and revoke privileges commands with various options.

12. Discuss on :

    (a) Data manipulation facilities.

    (b) Data Definition facilities.

13. Mention the benefits of ORACLE.

14. How to join multiple tables?

15. Explain the following :

    (a) Creating a table

    (b) Modifying a table

    (c) Dropping a table.

———————

DE-6742 **14**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

First Semester

OPERATING SYSTEMS

(2011 calendar year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is an operating system?

2. Define :

   (a) Process

   (b) Processor

   (c) Processes

3. With an example, discuss on mutual exclusion.

4. Explain the concept of two level scheduling.

5. Give the purpose of device controller.

6. Briefly explain the deadlock.

7. Give the usage of virtual memory.

8. What is meant by file system reliability?

PART B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Describe the structure of operating system.

10. Explain the concept of round robin scheduling.

11. Discuss on priority scheduling.

12. How to detect and recover from deadlock? Explain.

13. Discuss on multiprogramming with fixed partitions.

14. Describe the algorithm for page replacement.

15. Discuss on Disk Space Management.

DE–6743 **21**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2011 Calendar year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Compare and contrast C++ with other languages.

2. Mention the basic concepts of object oriented programming.

3. Explain the usage of IF and NESTED ..... IF statement in C++.

4. Give the purpose of Structures.

5. How to pass the arguments and objects in functions?

6. Discuss the execution of Inline functions.

7. Explain the concept of polymorphism.

8. How to open and close a file in C++?

PART B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Discuss the benefits of object oriented programming.

10. Write a C++ program to calculate the factorial of a given integer.

11. Explain the execution of SWITCH statement.

12. How pointers can be used in C++?

13. Write a C++ program to implement the concept of overloading functions.

14. Discuss the following inheritances :

    (a) Single.

    (b) Hierarchical.

    (c) Hybrid.

15. Write a program in C++ for updating a file.

DE–6744 **22**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

Second Semester

UNIX AND SHELL PROGRAMMING

(2011 Calendar year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the file and common commands in Unix.

2. Discuss about the command arguments and parameters.

3. Brief about the filters in Unix.

4. Explain on bugs and debugging in Unix.

5. Discuss about the functions and procedures in Unix.

6. Write a note on Unix system calls.

7. Brief the editors in Unix.

8. Explain the shell variables.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write a shell program to add ‘n’ numbers.

10. Explain various IS commands with an example.

11. Write a shell program to display all files in a given directory.

12. Discuss in detail about the various looping statements available in Unix.

13. Write a shell script to check the given string is palindrome or not.

14. Write a shell script to perform simple calculator.

15. Describe the I/O redirection in detail.

DE–6745 **23**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

Second Semester

COMMUNICATION SKILLS

(2011 Calendar Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write short notes on :

   (a) Objectives of Intra and Inter personal communication

   (b) Types and model of the communications.

2. Explain in detail Reading styles and Reading Techniques.

3. Discuss about

   (a) Importance and Advantages of Listening

   (b) Art of conversation and methods of asking questions.

4. Explain about process of Group discussion.

5. Explain about

   (a) Fundamentals of negotiation

   (b) Managing process of negotiation.

6. Explain about

   (a) Communication in management

   (b) Communication by specialists.

7. Explain about Employee's communication and Downward and Upward communication.

8. Discuss about techniques for delivering presentation and personel style.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe about the process of Effective writing skills in detail.

10. Write an essay about Interview Techniques in detail.

11. Discuss about the Body language and its tools in detail.

12. Discuss in detail about Group discussions.

13. Explain in detail Meetings and its objectives.

14. Describe about presentation planning, designing written and oral skills in detail.

15. Explain about communication in the management in detail.

DE–6746 **24**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

Second Semester

COMPUTER GRAPHICS

(2011 calendar year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. (a) Differentiate random scan and vector scan displays. (3)

   (b) Briefly describe any two input/output devices. (5)

2. Describe the Bresenham's algorithm for generating straight lines.

3. What is meant by composite transformation?

4. Explain the technique for clipping a point.

5. How to get a mirror reflection for a three dimensional object?

6. Discuss the concept of depth comparison.

7. Explain the concept of clipping against a finite view volume.

8. Mention the components of user interface. Explain.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe the algorithm for circle generation.

10. Explain the working style of a cathode ray tube for a video display device.

11. Discuss the following two dimensional transformations with examples.

    (a) Translation

    (b) Scaling.

12. Describe the Sutherland Hodgman algorithm for clipping.

13. How a three dimensional object can be rotated at origin and at any arbitrary point?

14. What is projection? Discuss its types with examples.

15. Discuss on the styles of command language.

DE–7607 **15**

DISTANCE EDUCATION

P.G.D.C.A. (S) DIPLOMA EXAMINATION, DECEMBER 2013.

First Semester

DATA STRUCTURES USING C LAB – I

(2011 Calendar year onwards)

Time : Three hours Maximum : 100 marks

One question should be given to each candidate by LOT system.

Each subdivision carries 50 marks

1. (a) Write a C program to find the all possible roots of a quadratic equation using function $ax^2 + bx + c = 0$.

   (b) Write a C program to implement push and pop operations on stack.

---------- Cut here ----------

2. (a) Write a C program to multiply any two order of matrix using function.

   (b) Write a C program for linked list implementation of Queue operations.

3. (a) Write a C program to the factorial of a given number using recursion.

   (b) Write a C program to implement insert and delete operations on Queue using array Concept.

---------- Cut here ----------

4. (a) A man is paid at the hourly rate of Rs. 15/- per hour for the first 45 hours worked. Thereafter, overtime is paid at 1.5 times the hourly rate for the next 25 hours and 2 times the hourly rate for further hours worked. Write a C program to input the no. of hours worked per week, calculation and print his gross weekly wage.

   (b) Write a C program to sort 10 Nos. in ascending order with name of variable and the value before and after sorting.

---------- Cut here ----------

5. (a) Write a C program to find the world is PALINDROME or not.

   (b) Write a C program for linked list implementation of Queue operations.

6. (a) Write a C program to read 10 values to an array variable. Use pointer to locate and display each value.

   (b) Write a C program to evaluate the given mathematical expression using stack.

---------- Cut here ----------

7. (a) Write C program to copy contents of one file to another file.

   (b) Write a C program to sort 10 Nos. in ascending order with name of variable and the value before and after sorting.

---------- Cut here ----------

8. (a) Write a C program to multiply 3×2 and 2×3 matrices of the order using function.

   (b) Write a C program for linked list implementation of Queue operations.

DE–7608 **16**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS
LAB – II

(2011 Calendar Year Onwards)

Time : Three hours Maximum : 100 marks

One question should be given to each candidate by lot system.

Break–up of marks.

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

1. Create a table "student" with the following details. Student name, Register No, Three test marks. Also write down the statements for the following.

   (a) Assign a letter grade to each student based on the average score, and List Register No, Average Score and grade.

Average Score	Grade
90–100	A
75–89	B
60–74	C
50–59	D
0–49	Fail

   (b) Count the number of students who got B grade.

   (c) List down the student details who get fail as the grade.

   The minimum pass for each subjects is 50.

2. Create table menu with fields item and unit – rate and create table item with tea, coffee and cool – drinks and create table bill with following fields. Empno, Empname, date of issue, item 1, no. of tokens issued for item 1, rate 1, item 2, no. of tokens issued for item 2, rate 2, item 3, no. of tokens issued for items 3, rate 3, total amount.

   Prepare a report with the following conditions.

   (a) Bill to customer

   (b) Display details on a particular date, the total sales amount

   (c) Display the total sales by an employee on a particular date.

---------- Cut here ----------

3. An airline reservation database contains the reservation table and personal table. Reservation table contains the following fields :

   Fight – No, class, passenger's name, seat no, personal table contains the following fields. Passenger name, Sex, Age, martial status, nationality.

   Write a PL/SQL statements to prepare the following reports.

   (a) Count the total number of married female passengers in a particular flight

   (b) Prepare a ticket with necessary details

   (c) Display the female passengers in the age group of 50 and above for all flights.

4. A daily sales file contains records with the following fields :

   Department number, date, item description, sale price for each item, quantity, cost of each item.

   Write a program using Oracle to list all the input data and compute total amount of sales and profit for each sale. Accumulate and print total of sales and profit. Show the output report in the following format.

   Department number, item description, sales price, quantity, cost, profit.

---------- Cut here ----------

5. ABC hospital maintains blood donors record information in a file. Create a table blood donor with following details, donorno, donor name, donor age, donor address, place of birth, blood group (A,B,O and AB).

   Also write down the queries for the following.

   (a) Blood donors having AB blood group.

   (b) Display the details of donors whose place of birth is chennai.

   (c) Count of number of donors whose blood group is A.

   (d) List the details of donors in the age group between 20 to 30.

DE–7609 **25**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

Second Semester

LAB – III – GRAPHICS USING C++

(2011 Calendar year onwards)

Time : Three hours Maximum : 100 marks

Record Note Book : 10
Algorithm : 10
Program : 50
Debugging : 10
Execution : 10
Result : 10
Total : 100

ONE question should be given to each candidate by LOT system.

1. Write a C++ program to do the following :

   (a) Create a base class shape which has the common properties of a circle and ellipse.

   (b) Inherit those properties to the child clanes circle and ellipse.

   (c) With the necessary methods draw those objects.

---------- Cut here ----------

2. Write a C++ program to draw a ship using necessary objects (line line, circle, arc).

3. Implement the Bresenham's line drawing algorithm using C++.

---------- Cut here ----------

4. (a) Write a C++ program to demonstrate the usage of Text graphics commands.

   (b) Draw a Circle, Rectangle, ellipse and fill it with a pattern.

---------- Cut here ----------

5. Write a C++ program which implements Bresenham's circle generating algorithm.

---------- Cut here ----------

6. Create a triangle and perform 2D transformations Translation and then scaling on the triangle object.

---------- Cut here ----------

7. Draw different lines by wearing a window and apply clipping algorithm on those line objects.

---------- Cut here ----------

8. Write a C++ program to manipulate the graphical object using 'shading' technique.

9. Implement the DDA line drawing algorithm using C++.

---------- Cut here ----------

10. Using C++ Graphics functions draw the symbol of olympics.

DE–7610 **26**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, DECEMBER 2013.

Second Semester

LAB IV — UNIX AND SHELL PROGRAMMING

(2011 Calendar year onwards)

Time : Three hours Maximum : 100 marks

Question should be selected by the Examiner and given to the candidates by Lot system.

Record : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. (a) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it :

   (i) File existence.

   (ii) File readable/writable.

   (iii) Both readable, writable

   (b) Write a shell program which accepts a file which contains list of names, and do the following :

   (i) Sort the names in a File.

   (ii) Search for a specific name.

   (iii) List out only top 5 names.

2. (a) Write a menu driven shell program to copy, edit, rename, delete a file.

   (b) Write a shell script which periodically monitors the disk for the existence of a file and then executes the program once the file has been located.

---------- Cut here ----------

3. (a) A hospital maintains patient details. Write a shell script to list the following :

   (i) Patients in a blood group wise.

   (ii) Patients in age between 20 and 30.

   (iii) Exit.

   (b) Write a shell script to generate leave status report of an organisation.

---------- Cut here ----------

4. (a) Write a script to prepare electric bill for domestic customers.

   For first 100 units – 0.75/unit

   Next 100 – 1.50/unit

   Above 200 – 3.00/unit

   Prepare the bill in the given format.

   Customer no name, Prev.Reading, IntReading units Amnt.

   (b) Write a script to send mail to groups of users by extracting their ids from /etc/group file.

5. (a) Write a script to print monthly pay slips for employee in an organization. Assume your own data and generate accordingly.

(b) Write a script using 3 arguments to take the pattern as well as input and output file names. If pattern not found display 'pattern not found' else display "Found" also check if right no of arguments are entered.